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Blockchain in smart education: Contributors, collaborations, applications and research topics

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Abstract
Researchers and practitioners are paying increasing attention to blockchain's potential for resolving trust, privacy, and transparency-related issues in smart education. Research on educational blockchain is also becoming an active field of research. Based on 206 studies published from 2017 to 2020, we identify contributors, collaborators, applications, and research topics using topic modeling and bibliometrics. Results indicate that currently, studies on the educational blockchain are mostly published in computer science conferences, with few in educational technology journals. Asian countries/regions (e.g., China and India) and institutions (e.g., University of Raharja) are active in this field. Countries/regions and institutions from the same regions tend to have close collaborations. Blockchain is mainly used to facilitate online testing and learning, education data mining and analytics, resource sharing, educational record verification and authentication, data management and storing, administration management, education/discipline of children, and college crowdfunding. Based on the results, we highlight the need to integrate artificial intelligence to enhance scalability and security and to facilitate justification and personalization of blockchain solutions based on accumulated big data to capture abnormal behaviors. This study assists researchers and practitioners in understanding educational blockchain research and becoming aware of the hot topics and future directions.

Keywords Smart education · Blockchain · Topic modeling · Bibliometrics · Contributors · Collaborations

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1 Introduction

Blockchain-based solutions, with features of decentralization, traceability, and immutability, are attracting attention to overcome challenges concerning trust, privacy, and transparency in smart education. The field of blockchain's application in education is increasingly active. There are reviews of the topic (e.g., Alammary et al., 2019; Bhaskar et al., 2020). However, many up-to-date articles were not included, and manual coding methodologies could be error-provoking. To present a timely review of blockchain's application in education, this study uses bibliometrics and topic modeling to enable a comprehensive, automatic, and objective analysis of up-to-date literature from the perspectives of contributors, collaborations, applications, and research topics.

1.1 Blockchain, features, and applications

A blockchain is a distributed, decentralized digital ledger where transactions and data are shared by a community in a verifiable and permanent manner and are secured through cryptography algorithms (Alammary et al., 2019). As part of the fourth industrial revolution, blockchain transforms "information sharing" into "value exchange" (Chen et al., 2018; Chung & Kim, 2016).

Powered by hash-driven proof-of-work distributed consensus algorithms, blockchain is characterized by decentralization, immutability, and authenticity (Ocheja et al., 2018). Specifically, a peer-to-peer decentralized, distributed architecture improves fault tolerance and information trust by avoiding centralized bottlenecks and failures (Veena et al., 2015). Furthermore, blockchain is immutable and secure by using a cryptographic hash where blocks on the chain cannot be erased or modified (Ocheja et al., 2019). Additionally, information is published and maintained collectively to ensure reliability and authenticity with consensus protocol ensuring that new blocks added to the chain are agreed upon by all participants (Chen et al., 2018).

Due to these capabilities, blockchain is revolutionizing data management and transaction, with increasing blockchain-based solutions being developed to resolve challenges about scalability, privacy, and reliability (Bhaskar et al., 2020). As estimated, by 2025, over 10% of global GDP will be stored on blockchains (Tapscott & Tapscott, 2017). Blockchain's various possibilities in diverse fields have caught considerable attention, for example, securing shared medical data, facilitating cross-border payments, and improving high education smartness. Education is one area where blockchain-based technologies trigger diverse opportunities.

1.2 Blockchain in smart education

In smart education, significant challenges include mobility, lifelong learning requirement, fraudulence, and the "Age of Data," where trust, privacy, and transparency in information sharing and retrieval are essential (Gong et al., 2018). Since the early days of educational technology, different guidelines, approaches, and policies are developed to resolve these challenges. Blockchain-based solutions, by providing extraordinary

features like decentralization, traceability, and immutability, show great potential to overcome the technical risks, possible threats, and privacy considerations in smart education (Hsu et al., 2022). Typical application scenarios are as follows.

First, distance learning opens up higher education to massive learners, which facilitates an increasingly mobile higher education (Arndt & Guercio, 2020). Alongside this heightened internationalization of education is a significant concern for securing data transactions (e.g., student profiles and certifications). Also, easy access to students' education records is vital to release friction and delays caused by multi-level administration. However, educational institutions commonly monopolize learning certification; thereby, students have no authority to access or manage it (Badr et al., 2019). Blockchain, by recording all cryptographic data transactions digitizedly, decentralizedly, and openly, allows students to become depositaries of their education records and effortlessly share them with interested parties (Hoy, 2017).

Second, alongside technological advances in the 21st Century workforce is a demand for skill updates for lifelong learning (Ocheja et al., 2019). However, the digitalization and globalization of learning environments (e.g., online learning) make it difficult and time-consuming to check the authenticity of one's diplomas and certificates attesting his or her skills and achievements. Blockchain provides capabilities to integrate education records from different educational providers for instant authenticity verification (Alammary et al., 2019). Also, with smart contracts as a traceable digital transaction facilitator, Blockchain addresses the gaps prevailing in copyright and credentials, which is promising for virtual and online learning (Guo et al., 2019).

Additionally, according to Risk Advisory Group,[1] 80% of CVs contained one or more discrepancies related mostly to educational qualifications. However, degree and diploma recognition, particularly those issued by foreign educational institutions, is challenging (Trines, 2017). Blockchain prevents fraud by managing educational data and data access to third parties. The reliability and authority "are both ensured to significantly reduce degree fraud since the data matched with users' ID and stored in blockchain are checked, validated, and maintained collectively (Chen et al., 2018, p.5)."

Due to the above advantages, various blockchain solutions are integrated into smart education systems. For example, Kamišalić et al. (2019) revolved around online education to develop smart contract-based systems. According to the Gartner 2019 CIO Survey, blockchain has already been deployed by 2% of education respondents, and 18% were planning to do so in the next two years.[2]

2 Research aims and questions

Compared to the blockchain as a purely technical concept (i.e., known as distributed ledger technology), educational blockchain concerns blockchain's application in smart educational environments to allow the systems to be more secure, reliable, and transparent (Hsu et al., 2022).

[1] https://www.riskadvisory.com/campaigns/cv-lies-2017/

[2] https://www.gartner.com/smarterwithgartner/4-ways-blockchain-will-transform-higher-education/

The exceptional opportunity of educational blockchain is increasingly discussed among researchers and practitioners. Focusing on blockchain's application in education, Grech and Camilleri highlighted blockchain's potential in certificate issuing, accreditation pathway verification, lifelong learning passports, and intellectual property management.[3] Chen et al. (2018) indicated blockchain's popularity in managing academic degrees and supporting summative evaluation for learning outcomes, whereas Al Harthy et al. (2019) indicates blockchain's effectiveness for data transaction, securing student records, digital badges, human resource, libraries, and publishing in higher education.

Several reviews analyze scientific studies concerning educational blockchain. Bhaskar et al. (2020) systematically analyzed 36 articles from Scopus to draw insights into the development, benefits, challenges, and applications of educational blockchain, indicating blockchain's popularity in certificate management and digital guardianship consent. However, scalability, security, and cost remained barriers. Alammary et al. (2019) systematically reviewed 31 articles to understand educational blockchain's applications, benefits, and challenges, indicating blockchain's popularity "to issue and verify academic certificates, share students' competencies and learning achievements, and evaluate their professional ability (p.16)." The benefits of educational blockchain included providing secure platforms for student data sharing, lowering costs, and promoting trust and transparency. Challenges concerning blockchain's application in education are "related to security, privacy, cost, scalability, and availability before blockchain adoption (p.16)."

The above reviews unfold the following (research) gaps. First, limited studies were reviewed, that is, around 30. Most reviews (e.g., Alammary et al.; Bhaskar et al.) claim that the field is in its infancy with limited work available; however, this limitation is also attributed to the limited sources they used. Bhaskar et al. only considered Scopus, whereas in Alammary et al., although more databases were used, the search was conducted in April 2019. As scientific advances happen rapidly, especially for such a field with great potential as educational blockchain, many up-to-date articles are now available. Additionally, to guarantee a timely review with full coverage of relevant literature and comprehensive analysis, it is more suitable to search Google Scholar, which was not used in previous reviews. Additionally, extant reviews relying on human coders' expertise suffer from error-provoking and coding inconsistencies. Thus, advanced technologies suitable for big data mining are expected to enable automatic and objective analysis. This assists policymakers, educational institutions, managers, and scholars in obtaining a detailed and comprehensive understanding of educational blockchain, which is useful in their research design and project planning.

With the increasing applications of educational blockchain alongside accumulative scientific studies, scholars and practitioners need to understand the present state-of-the-art practices. Accordingly, this study analyzes the most recent literature on the educational blockchain using bibliometrics and topic modeling. Specifically, we focus on contributors, collaborations, applications, and research topics. This

[3] https://www.pedocs.de/volltexte/2018/15013/pdf/Grech_Camilleri_2017_Blockchain_in_Education.pdf

study contributes to educational blockchain in extant scenarios and future applications. There are three research questions.

RQ1: What were the top publication sources, countries/regions, and institutions?
RQ2: What were the scientific collaborations?
RQ3: What were the most investigated research topics?

3 Dataset and methods

3.1 Data preparation

Figure 1 shows the data collection and screening. Web of Science, Scopus, Education Resources Information Center, and Google Scholar were searched using terms related to the blockchain ("block chain*" and "blockchain*") and education ("education*," "learning," and "teaching") on January 22, 2021. A total of 1578 papers were collected. A duplication process based on title similarity deleted 90 papers, leaving 1488 papers for full-text screening. Given a paper, we started from the first criterion (i.e., relevant to education). If it was not, we excluded it directly. Totally, 1071 papers were excluded. Subsequently, we checked whether it was about blockchain and included details concerning its application in education, thereby excluding 21 papers. After confirming that

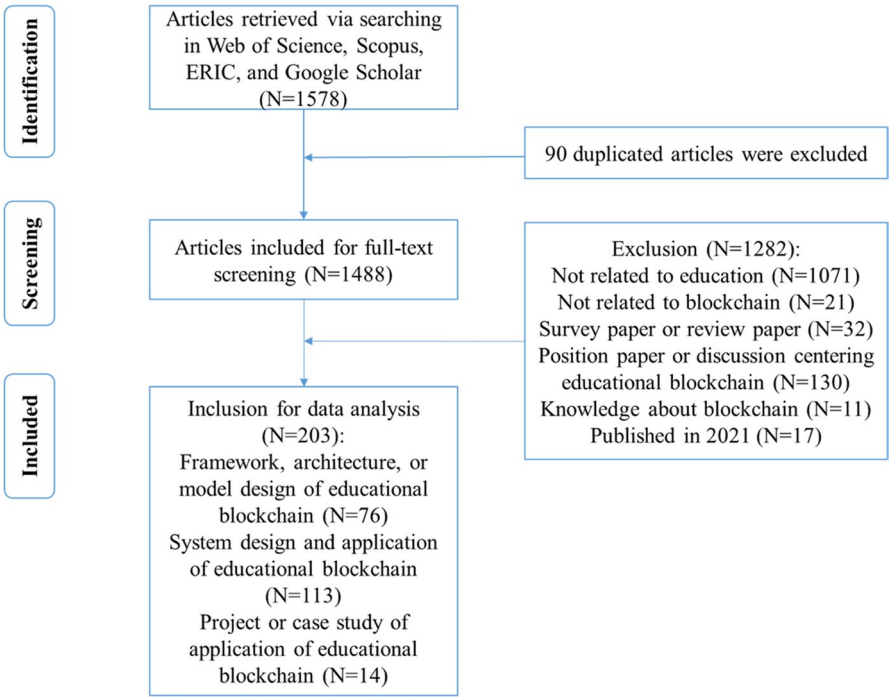
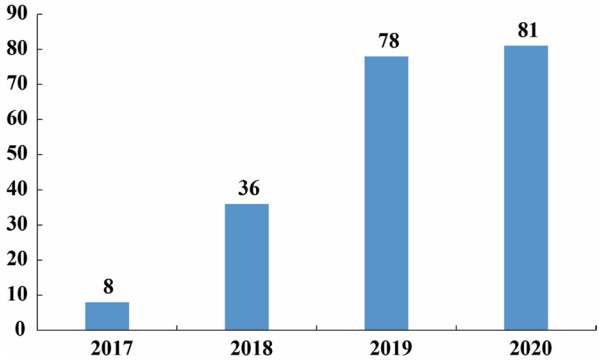


Fig. 1 Data collection and screening

Fig. 2 Number of papers



the paper was about educational blockchain, we evaluated its originality and excluded 32 reviews and surveys accordingly. Furthermore, 130 position papers on educational blockchain's potential and 11 papers introducing knowledge about blockchain were excluded. Additionally, 17 papers published in 2021 were excluded. The screening resulted in 203 relevant articles that were categorized into 1) framework, architecture, or model design of educational blockchain, 2) system design and application of educational blockchain, and 3) project or case study of the educational blockchain. Figure 2 shows the annual distribution of the 203 articles. The data displayed in the figure was determined by counting the articles published in each given year. Overall, starting from 2017, blockchain's applications in education have received growing attention among scholars. The details of the 203 articles can be accessed via https://docs.google.com/spreadsheets/d/117Q9lUddp3SqfntuYJ_-OeOY5PaI7ID4/edit?usp=sharing&ouid=117809493554014724690&rtpof=true&sd=true.

3.2 Data analysis

The 203 studies were analyzed in terms of contributors, collaboration, applications, and research topics. Bibliometric indicators, social network analysis, term frequency analysis, and structural topic modeling (STM) (Roberts et al., 2014a, 2014b) were used.

3.2.1 Bibliometric indicators and social network analysis

To answer RQ1, this study adopted bibliometric indicators involving the number of articles (*TA*), the number of citations (*TC*), the Hirsch index (H-index), and the average number of citations (*ACP*) to discover active publication sources, countries/regions, and institutions. Specifically, the numbers of articles and citations were utilized for measuring the production and impact of publication sources, countries/regions, and institutions. *TA* a publication source, country/region, or institution (represented by *i*) was calculated by totaling the number of its contributing articles, and *TC* was the sum of the citations received by its contributing article. *ACP* of a publication source, country/region, or institution was computed using Eq. (1). The H-index, which indicates that h of an individual's publications has at least h citations each, was utilized to evaluate the academic level of a publication source, country/region, or institution from the qualitative and quantitative perspectives.

$$ACP_i = \frac{TC_i}{TA_i} \tag{1}$$

To answer RQ2, social network analysis was utilized to visualize the collaborations among institutions and countries/regions via Gephi (Bastian et al., 2009). Each institution or country/region is illustrated by a node with its size indicating its production (i.e., the number of its contributing articles). A link connecting two nodes indicates that their represented institutions or countries/regions have collaborations, with its width indicating their collaborative frequencies (i.e., the number of their collaborative articles). The colors of nodes suggest the located continents of a country/region or the located country/region of an institution.

3.2.2 Keyword analysis and structural topic modeling

To answer RQ3, frequencies of phrases were calculated and sorted to pinpoint the frequently researched topics. We further tracked the history of the field of educational blockchain by averaging the publication years of the articles that mention each phrase to detect emergent research topics.

We also conducted topic modeling analysis to pinpoint the intellectual structure and underlying topics within the educational blockchain literature. STM, as a complex probabilistic topic modeling technology, has its plate diagram shown in (Roberts et al., 2014b) Fig. 3.

Given a document corpus indexed by D with vector x_d indicating document-level covariates, the number of topics K, total n words in a sampled document d indicated by $\{w_{d,n}\}$, and vocabulary size V, a basic pseudocode of the STM modeling process is described as follows (Roberts et al., 2016).

The first step focuses on estimating the topic prevalence parameter θ_d for each word in document d through logistic-normal generalized linear modeling. θ_d is created using Eq. (2) for each word from V vocabularies with the possibility of $k = 1, \ldots, K$ topics. In the equation, the topic prevalence model's coefficients are indicated by $\Gamma = [\gamma_1 | ... | \gamma_K]$, with Σ being a $(K-1)$ by (K-1) covariance matrix.

$$\theta_d = \text{LogisticNormal}_{k-1}\left(\Gamma' x_d, \Sigma\right) \tag{2}$$

The second step concentrates on estimating the topical content model β to represent the words as a probabilistic mixture of each topic (k) using Eq. (3), in which m stands for the standard word distribution vector of length V, $k_{k,v}^{(t)}$ refers to the topic (t) specific deviation, $k_{y_d,v}^{(c)}$ indicates the covariate (c) group deviation, and $k_{y_d,k,v}^{(i)}$ suggests the interaction (i) coefficient.

$$\beta_{d,k,v} = \frac{\exp(m_v + k_{k,v}^{(t)} + k_{y_d,v}^{(c)} + k_{y_d,k,v}^{(i)})}{\sum_v \exp(m_v + k_{k,v}^{(t)} + k_{y_d,v}^{(c)} + k_{y_d,k,v}^{(i)})} \tag{3}$$

Third, for each word in the document, ($n \in \{1, \ldots, N_d\}$), a topic is sampled using a multinomial distribution over θ_d, shown in Eq. (4); for a given topic, a word is

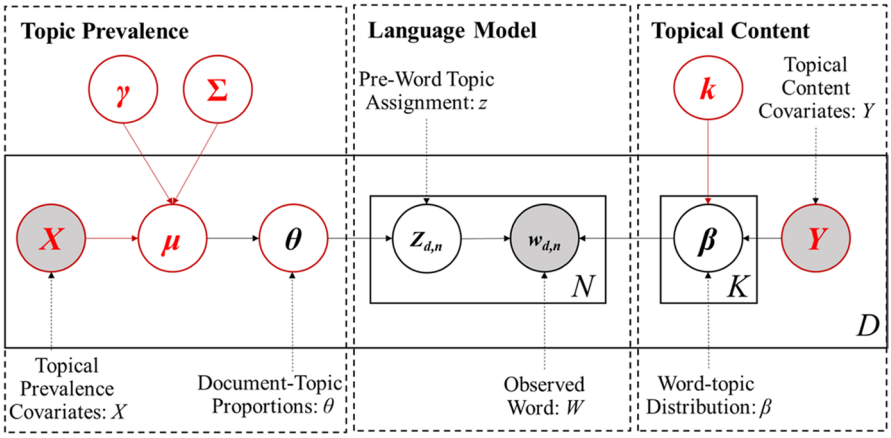


Fig. 3 Plate diagram of STM (Adapted from Roberts et al. (2014b))

sampled using Eq. (5), in which B refers to a global parameter associated directly with document-level covariates.

$$z_{d,n} \sim \text{Multinomial}_K\left(\theta_d\right), \text{ for } n = 1, \ldots, N_d \tag{4}$$

$$w_{d,n} \sim \text{Multinomial}_K(\text{B}, z_{d,n}), \text{ for } n = 1, \ldots, N_d \tag{5}$$

In this study, the STM was achieved using an R package *stm* (Roberts et al., 2014a). Before modeling, we used Term Frequency-Inverse Document Frequencies (TFIDF) to quantify each word's comparative importance in the corpus. In TF-IDF, Term frequency (TF) assesses the frequency of occurrence of a word. Given D documents and V unique words in a corpus, the TF value (TF_{ij}) of the ith word in the jth document is determined using Eq. (6), in which C_{ij} and N_j represent the number of times the ith word appears and the total number of words in the jth document.

$$TF_{ij} = \frac{C_{ij}}{N_j} \tag{6}$$

Inverse Document Frequency (IDF), measuring a term's importance, is computing using Eq. (7), in which IDF_i is the ith word's IDF value and D_i is the number of documents mentioning the ith word. The ith word's TF value in the corpus is determined using Eq. (8).

$$IDF_i = \log_2(\frac{D}{D_i}) \tag{7}$$

$$TF_i = \sum_{j=1}^{D} TF_{ij} * IDF_i \tag{8}$$

All words are sorted by their TF-IDF values, with a TF-IDF threshold setting as 0.03 to remove words with a TF-IDF value lower than the threshold. The remaining words were used as data input in the topic modeling. A multiple modeling strategy (Chen et al., 2021, 2022) indicates that the model with 11 topics has the greatest semantic consistency (Silge, 2018) and exclusivity (Kuhn, 2018). Based upon the estimated θ, the prevalence of each topic is determined using Eq. (9), where P_k is the prevalence of kth topic in the corpus and $\theta_{d,k}$ $(d = 1, ..., 203, k = 1, ..., 10)$ is the probability of document d associated with the kth topic.

$$P_k = \frac{\sum_d \theta_{d,k}}{D} \tag{9}$$

4 Results

4.1 Publication sources, countries/regions, and institutions

The 203 articles were published in 173 sources. Among the top eight (Fig. 4), only two were journals, whereas most were conferences, with International Conference on Advanced Learning Technologies and International Conference on Education, Research and Innovation as the top two. IEEE Access, one of the only two journals, had the most citations, although with only two articles. This indicates a higher scientific impact of journal articles on educational blockchain compared to conference papers.

There were 65 countries/regions and 283 institutions. Among the top 13 productive countries/regions (Fig. 5) and top 14 productive institutions (Fig. 6), China and the University of Raharja showed the best performance.

4.2 Scientific collaborations

The collaborations among ten productive countries/regions and 13 institutions are shown (Figs. 7 and 8). Regarding countries/regions, there are four groups of collaborators. France and Tunisia collaborated in publishing three articles, and the remaining collaborators collaborated in publishing two research articles. Regarding institutions, there are five groups of collaborators. The collaborations among institutions from the same regions are close, for example, the University of Reims and the University of Paris 08. There are four pairs that have collaborated in publishing three articles, including the University of Paris 08 and University of Reims Champagne-Ardenne, NPO CCC-TIES and Kumamoto University, NPO CCC-TIES and Kyoto Women's University, and Kyoto Women's University and Kyoto Women's University. The remaining collaborators collaborated in publishing two research articles.

4.3 Frequent and emerging phrases

Table 1 presents the frequently used phrases, with the integration of the timeline averaged by the publication years of the papers mentioning each phrase. Table 2 presents

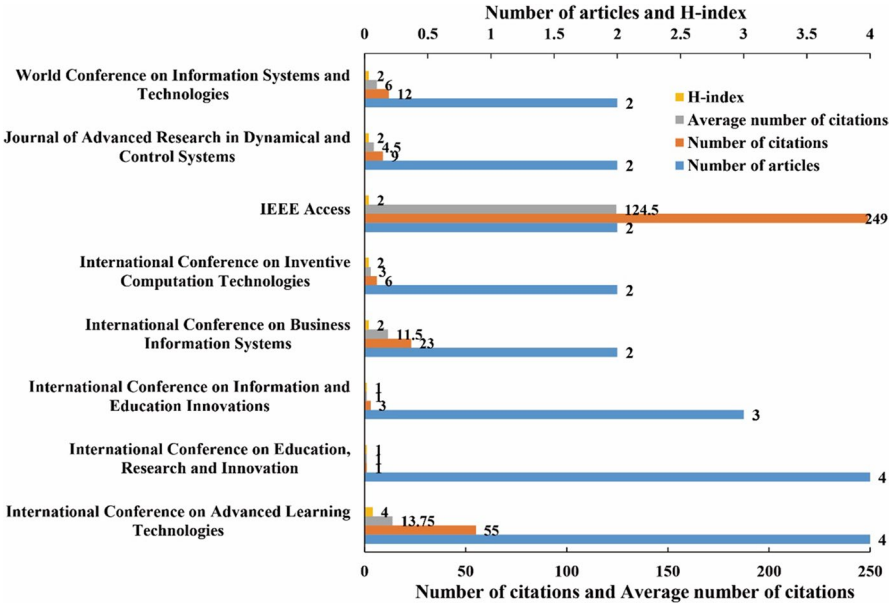


Fig. 4 Top publication sources

emergent phrases along with their frequencies. The results show blockchain's popularity in online learning, lifelong learning, and achievement data management. Additionally, big data, fake certification, private data, collective maintenance, secure storage, and adaptive education are increasingly concerned. Technologically speaking, learning analytics (LA), network technologies, and artificial intelligence (AI) gradually appeared.

4.4 Research topics

Figure 9 presents topic modeling results with 11 topics. The most popular topics included certificate verification and authentication, learning outcome and awarding, educational resource management, and data storage and management.

5 Discussion

5.1 Answers to research questions

This topic-based bibliometric study analyzes contributors, collaborations, applications, and research topics in educational blockchain research. Since 2017, this field has gained increasing attention, particularly in the recent two years. Results to RQ1 are consistent with Alammary et al. (2019), indicating that, currently, most studies on the educational blockchain are published in computer science conferences, whereas fewer are in educational technology journals. We thus call for international journals to launch relevant

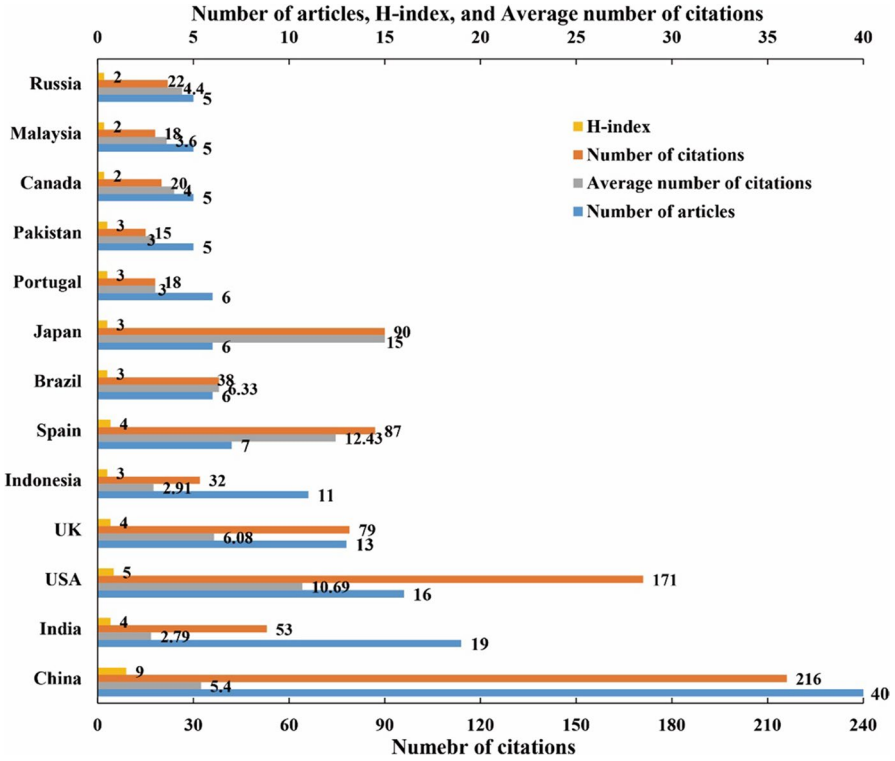


Fig. 5 Top countries/regions

special issues to encourage more advanced work in this field. Similar to Alammary et al., we indicate active countries/regions (e.g., China and India) and institutions (e.g., the University of Raharja) from Asia. This is perhaps because of the great enthusiasm for cryptocurrencies among Asian countries/regions where there are many of the world's largest cryptocurrency exchanges (Bank, 2018). Results to RQ2 indicate the close collaborations among countries/regions and institutions located in the same regions, whereas cross-regional collaborations should be enhanced.

Results of RQ3 indicate similarities and differences in research topics compared to previous reviews (Alammary et al., 2019; Bhaskar et al., 2020). Bhaskar et al. and Alammary et al. mentioned certificate and degree management, while we indicate blockchain's prevalence in certificate, credit, degree verification, authentication, and anti-forgery. Bhaskar et al. and Alammary et al. highlight examination, online education, and collaborative learning as three separate topics, while we highlight an increasingly promising field of online testing and online collaborative learning that is facilitated by online quizzes' popularity to allow quick updates for quiz creation and the prevalence of collaboration in online learning mode in recent years (Tentea et al., 2019). Bhaskar et al. and Alammary et al. mentioned learning outcome management, while we emphasize the management of all types of achievement records. Bhaskar et al. and Alammary et al. identified lifelong learning, whereas we highlight blockchain in data management and storing of competence, capability, and professional

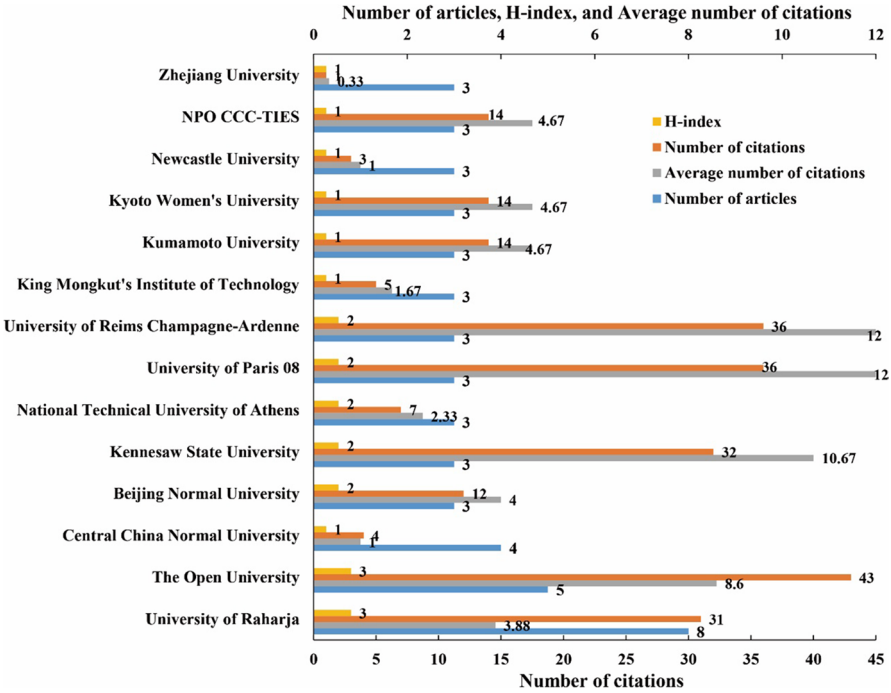


Fig. 6 Top institutions

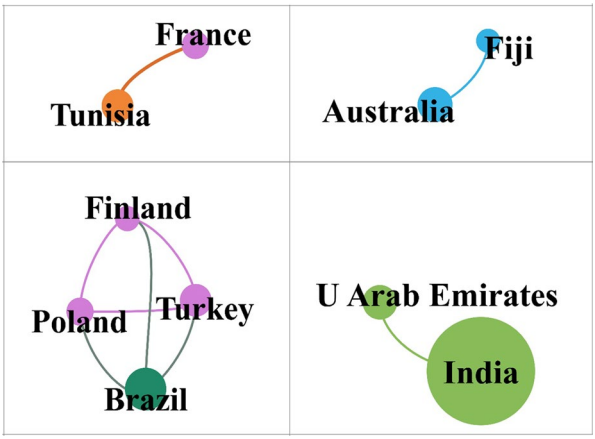


Fig. 7 Collaborations among countries/regions

ability evaluation for lifelong learning. Bhaskar et al. and Alammary et al. emphasized copyright management, digital guardianship consent, and fee and credit transfer, which are covered in digital asset and funding management and data and resource sharing in our results. Although our results reflect almost all issues mentioned in previous reviews, we do not identify competition management as Bhaskar et al. and Alammary et al. did. This is maybe because of its relatively lower occurrence in the

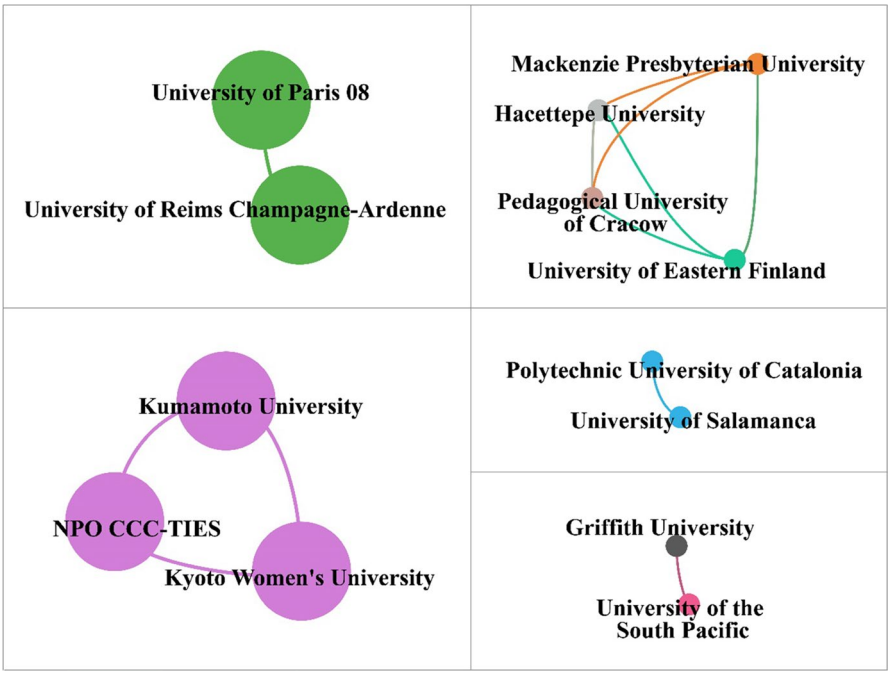


Fig. 8 Collaborations among institutions

data corpus since we mainly focus on frequently-discussed topics. Another possible explanation is the topic overlaps in topic models, which may cause its coverage into topics like data management and storing. Nevertheless, as we have a broader coverage of up-to-date studies, our results are more comprehensive. We additionally highlight issues not identified previously, for example, blockchain for curriculum evaluation, awarding, adaptive education, background screening, LA, blockchain learning, AI, and network technologies integrated into blockchain educational systems.

Subsequently, instead of repeating what is available in previous reviews (e.g., document verification and authentication), we grouped topics into seven themes for discussion. For example, considering blockchain's importance for online learning examination and online collaborative learning evaluation, we form a theme of blockchain-integrated online testing. The same is true of other themes except for the last one which includes multiple scenarios like operational efficiency, curriculum evaluation, education/discipline of children, and crowdfunding tertiary education.

5.2 Important research topics

5.2.1 Blockchain-enhanced LA

Blockchain benefits LA by ensuring data privacy and security and providing transparent and reliable data to promote users' confidence in LA. Recently, alongside the

Table 1 Top frequent phrases

Phrases	Timing averaged by the publication years	Frequencies (Number of papers)	Phrases	Timing averaged by the publication years	Frequencies (Number of papers)
smart contract	2019.3	51	information technology	2019.3	4
blockchain-based system	2019.2	27	blockchain-based certification system	2019.8	4
educational institution	2019.3	23	digital education certificate	2019.8	4
third party	2018.9	22	issuing institution	2019.5	4
educational system	2019.4	20	secure storage	2019.5	4
online education	2019.5	13	digital signature	2019.3	4
blockchain network	2019.3	12	industrial revolution	2019.3	4
digital certificate	2018.9	12	learning process	2019.3	4
distributed ledger	2019.6	10	teaching platform	2019.3	4
blockchain platform	2019.4	10	decentralized way	2019.3	4
academic certificate	2019.2	10	private blockchain	2019.3	4
educational certificate	2019.4	9	verification process	2019.3	4
artificial intelligence	2019.8	8	academic institution	2019.0	4
academic credential	2019.5	8	computer science	2019.0	4
ethereum blockchain technology	2019.4	8	digital technology	2019.0	4
education institution	2019.1	8	distance learning	2019.0	4
permissioned blockchain	2019.1	7	learning record	2019.0	4
learning activity	2018.9	7	lifelong learning	2019.0	4
blockchain-based solution	2019.5	6	tertiary education	2019.0	4
case study	2019.0	6	open online course	2018.5	4
verification system	2019.3	6	ubiquitous environment	2018.5	4

Table 1 (continued)

Phrases	Timing averaged by the publication years	Frequencies (Number of papers)	Phrases	Timing averaged by the publication years	Frequencies (Number of papers)
learning analytics	2018.3	6	blockchain model	2018.8	4
educational resource	2019.6	5	learning model	2018.8	4
large number	2019.4	5	open badge	2018.5	4
issued certificate	2019.2	5	distributed system	2018.5	4
learning achievement	2018.8	5	public ledger	2018.5	4
smart e-learning system	2019.5	4	learning management system	2017.8	4

Table 2 Key emerging phrases in 2020

Phrases	Frequencies	Phrases	Frequencies	Phrases	Frequencies
private data	3	anti-forgery mechanism	1	identity authentication	1
decentralized model	3	attribute-based encryption	1	information self-verification	1
education resource	3	augmented reality	1	instant certificate verification	1
fake certificate	3	automated credit accreditation	1	interactive data authentication	1
online teaching	3	award system	1	iot system	1
peer-to-peer environment	3	background screening	1	language learning	1
consensus algorithm	3	blockchain-enabled iot	1	lifelong competency	1
data storage	3	bloxxgame tool	1	long term storage space	1
decentralized application	3	bulk submission	1	medical education credit system	1
data management	2	campus governance	1	mobile higher-education	1
achievement record system	2	chain-based identity management	1	network analysis technology	1
big data	2	collaborative processing	1	neural network	1
blockchain-based authentication system	2	collective maintenance feature	1	off-chain capability	1
certification authority	2	competency profile	1	peer feedback	1
cost reduction	2	continuing education	1	personalized recommendation	1
data collection	2	cost-effective verification system	1	personalized service platform	1
digital diploma	2	cross-acquired certificate	1	pervasive use case	1
digital education transaction	2	cross-university collaboration	1	pow algorithm	1
education certificate verification	2	curriculum personalisation	1	process management	1
education information system	2	customized verification	1	professional profile	1
education student certificate	2	data transmission	1	project-based learning	1
english teaching	2	data transparency	1	prospective job applicant	1
future learning	2	data verification	1	quick online reference	1
information processing	2	data-driven culture	1	rbft algorithm	1

Table 2 (continued)

Phrases	Frequencies	Phrases	Frequencies	Phrases	Frequencies
interplanetary file system	2	decentralized governance	1	real-time testing	1
management system	2	decentralized peer-to-peer infrastructure	1	repairing management	1
network technology	2	digital asset	1	responsive learning environment	1
open learning	2	resource authentication	1	revocation policy	1
qr code	2	digital identity authentication	1	revolutionary blockchain technology	1
revocation mechanism	2	digital right management system	1	scholarship management platform	1
tamper-proof ledger	2	diploma forgery	1	secure collaborative learning system	1
whole system	2	diploma fraud	1	self-regulated learning	1
lifelong education	2	distributed computing	1	semantic blockchain	1
multimedia learning resource	2	document integrity	1	sharable student credential	1
object authentication	2	dynamic authorization code	1	shareable education material	1
online verification	2	forgery protection	1	storage security	1
academic credit transfer	1	fuzzy information clustering	1	student model	1
account-based blockchain model	1	gamified blockchain learning app	1	transaction-based blockchain	1
adaptive education	1	genetic algorithm	1	virtual classroom	1
alleviate fake diploma problem	1	global education system	1	web-based simulation game	1
analogue operation	1	human intelligence	1	whiteboard-based blockchain game	1

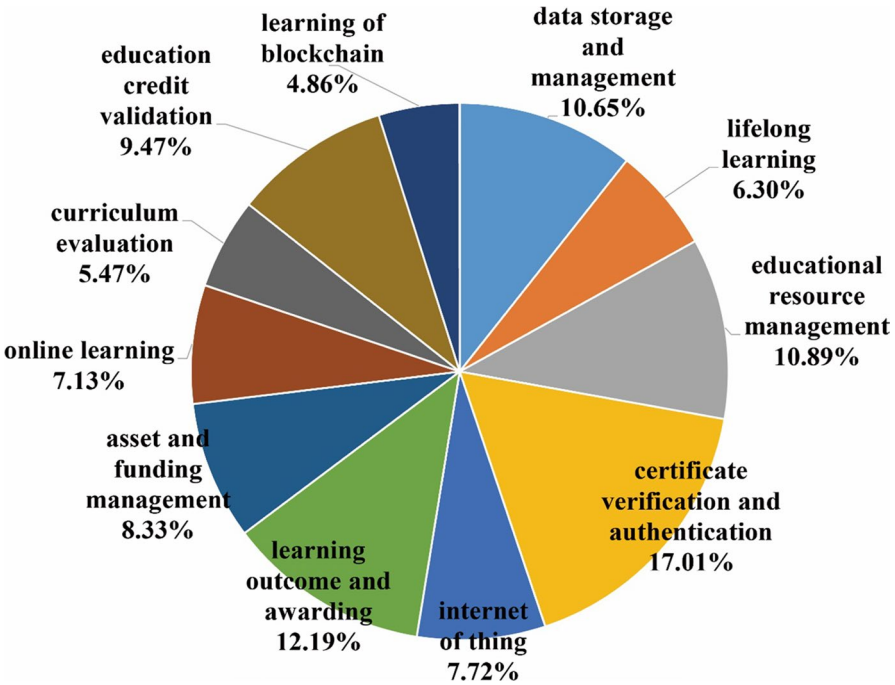


Fig. 9 Research topics and proportions

growing interest in LA is a concern for data ownership among institutions and students. While LA contributes to enhancing instruction and learning quality, students' sensitive data collection brings privacy and security problems and decreases confidence in their use. Privacy and identity management to ensure student agency over data in LA appears essential (Filvà et al., 2018). However, currently, laws and policies fail to fully guarantee students' data privacy, security, validity, integrity, and reliability.

Blockchain provides technological solutions to LA ethics and trust problems. Specifically, data about students' learning activities used in LA is stored in decentralized distributed ledgers to automate rules and constraints to allow student governance, thus ensuring data privacy and security (Filvà et al., 2018). Hillman et al. (2019) adopt an immutable and publicly verifiable data management system to assign learners and colleges an immutable log alongside full access to data stored in systems, whereas third parties that request access to college data is assigned a unique virtual token to track data modifications, access, and use. All the involved users' various interactions with data are notarized and timestamped by smart contracts, and participants in online learning activities can store learning traces in any place they like. Further, with smart contracts, stakeholders retrieve and securely share original data with third parties, thus offering a transparent and reliable data source for LA. Additionally, blockchain coupled with other technologies facilitates learning and instruction. An example is blockchain's combination with digital storytelling (Oyelere et al., 2020) for providing students with smart, accessible, and responsive learning experiences. Besides assisting instructors in analyzing students'

progress via LA and reflecting on their instructional practices, blockchain helps students access learning content and resources during digital storytelling activities without time and space constraints.

5.2.2 Blockchain-integrated online testing

Formative assessment has difficulty in tracking every detail during evaluation, particularly in e-learning, where the commonly used online quizzes bring troubles in irregularities and test result storage. Blockchain and smart contracts handle this challenge by allowing single sign-on and custom quiz creation and preventing quizzes and result tampering to ensure authentic and antitheft assessments. When students submit work to blockchain-based platforms, smart contracts review students' performance and record the results into blocks. By coupling with genetic algorithms to analyze examination processes, the systems complete various tasks like question generation and result evaluation and verify online testing, thus allowing more intelligent, standardized, and highly efficient examination and preventing students' irregularities (Wu, 2020). Such advantages also apply to online collaborative learning evaluation by saving behaviors and dialogues that happened during collaborations into blocks as a basis for performance evaluation and considering students' opinions during assessments to solve the free-riding problem and ensure evaluation fairness (Chen et al., 2018).

5.2.3 Data and resource sharing

The major challenges of data sharing involve high costs, user identity, data trade, exchange management, and data security (Lu, 2019). Blockchain-based systems combined with storage servers and cryptography techniques allow low-cost and reliable sharing across education systems and institutions (Ocheja et al., 2018). Specifically, ledgers track and manage data transactions between multiple parties in real-time, and reduce management costs during data sharing through access rights to ensure data reliability and confidentiality. Further, original educational records stored in encrypted form in off-chain storage servers are anchored periodically with the hash information on the blockchain to ensure data security. Additionally, smart contracts regulate the storage and sharing processes, and cryptography techniques handle records encryption and digital message signature (Lu, 2019). This facilitates university identity authentication and collaborative processing of educational resources among universities.

Application scenarios of such systems include credit transfer, test question sharing, and interdisciplinary talent cultivation. In Srivastava et al. (2018), a globally trusted blockchain-driven educational framework verifies students' academic certificates and course credits and digitally transfers them among various stakeholders (e.g., universities and companies). This ensures an inclusive view of students' performance by maintaining consistency among local copies of educational certificates and credits. According to Badr et al. (2019), such a system with reliable verification reduces wait times for students' record transfer and avoids academic fraud. Islam

et al. (2019) focus on question sharing by encrypting a new question paper using timestamp and salt hash and hashes from previous encrypted question papers and storing it in blockchain with smart contracts to allow students to unlock selected question papers. A timestamp-based lock further restricts beforehand decryption of question papers, thus ensuring test confidentiality. According to Chen et al. (2019), students' digital resumes on "online learning experiences, project outcomes, and certification of completion" are shared among institutions, schools, and companies to "monitor and manage performance and training history" via interdisciplinary talent cultivation by linking "resources in academic and industrial fields (p.113)."

5.2.4 Blockchain-facilitated online learning

Alongside distance education's prevalence is modern technologies' use in smart e-learning systems to trace student activities. However, data privacy and security remain challenging. Blockchain is promising to facilitate multimedia resource management, digital identity authentication, payment of online courses, and online certificate verification. Specifically, a blockchain-based digital right management system facilitates multimedia learning resource sharing and management using combined public and private blockchains, where smart contracts record multimedia digital rights, secure storage, and unmediated digital certificate verification (Guo et al., 2019). Blockchain guarantees identity data security and reliability in digital identity authentication by storing identity information in a ciphertext manner. Zhang et al. (2020b) utilize "a trusted bulletin board deployed on blockchain" to allow effective "privacy protection in collaborative learning" by guaranteeing "authentic and correct message interaction (p.47)." By using smart contracts, learners achieve consensus to restrain malicious behaviors, which regulates participants' behaviors during collaboration. According to de Castro et al. (2017), blockchain enables learners to learn in their own time and schedule reinforced by social interactions and federated access and to share multimedia materials securely. Furthermore, to ensure security and resolve inefficiencies, composer smart contracts and network topologies based on permissioned blockchain allow fast payment and trusted audit in online learning (Lu et al., 2019). Additionally, permissioned blockchain, coupled with digest algorithms and interactive data authentication via digital signatures, facilitates digital certificate issuing with instant verification between learners and third parties through QR codes (Cheng et al., 2020).

5.2.5 Data management and storage

Different types of student documents (e.g., transcripts, achievements, and scholarship records) are stored, managed, and protected using blockchain. Benefits include security, preserving privacy, and allowing long-term storage for lifelong learning (Nurhaeni et al., 2020). UniChain (Daraghmi et al., 2019), a blockchain-grounded system, provides students, universities, and third parties with interoperable, secure, and effective access to electronic academic records while keeping student privacy. UniChain adopts timed-based smart contracts to manage transactions and monitor

access and advanced encryption techniques to provide security. In Āriņš (2019), learners "publish evidence of their learning achievements in desired formats connected with blockchain for instant authenticity verification," thereby eliminating "knowledge passport fraud and reducing organizational workload overhead for achievement verification (p.207)." Blockchain also supports decentralized data management in the Internet of Things (IoT) systems, where IoT learning-assisted resources are stored in the blockchain for future allocation. Alam and Benaida (2020) combined "blockchain and IoT to create an interaction system between students, teachers, employers, developers, facilitators, and accreditors on the Internet (p.2164)."

5.2.6 Digital games for Blockchain learning

Alongside educational blockchain's prevalence is an increasing interest in blockchain learning. Among diverse instructional practices, gameplay experiences where players learn by accomplishing game tasks virtually facilitate blockchain learning. In flipped learning (Weng et al., 2019), a role-playing game with three-dimensional animation and comic teaching materials is used to teach blockchain knowledge in task clearance modes. Bloxxgame (Dettling & Schneider, 2020), a web-based simulation game, supports experience-based blockchain concept instruction by enabling students to experiment with relevant operations like signing, creating transactions, and building blocks. Additionally, project-based learning, coupled with self-regulated learning and peer feedback, facilitates blockchain learning by asking students to write research papers or work in groups on programming projects to build and deploy blockchain systems written in Solidity language (Mentzer et al., 2020).

5.2.7 Blockchain in other contexts

Blockchain is also adopted in administration, college curriculum evaluation, education/discipline of children, and crowdfunding tertiary education. Blockchain facilitates administration through institution management and education resource extraction. An example is Vasiuta and Buniak (2019), where users create and execute smart contracts into workflow systems in higher education institutions. Because no centralized or trusted parties are required, it significantly reduces the number of paper-based information carriers and automates the existing workflow systems. Zhang et al. (2020a) focus on high-quality education resource extraction to improve educational administration via optimal scheduling and educational resource acquisition based on blockchain trust and spatial distribution resource scheduling models. Specifically, statistical analysis is utilized to understand the convergence of high-quality education resource extraction processes. Then, quantitative feature analysis and fuzzy information clustering are used to extract and control these resources. Additionally, the root game equilibrium optimization algorithm optimizes resource quality. To resolve the drawbacks of non-transparence and centralization in the college curriculum, a blockchain-driven solution (Wu et al., 2019) separates curriculum evaluation from college educational systems and adopts blockchain to achieve

evaluation result validity and openness. Furthermore, blockchain facilitates the education or discipline of children at home via parental control schemes based on tokens where parents encourage children's good behaviors and depress ill ones via tokens transferred on the blockchain (Suchaad et al., 2018). Specifically, the control scheme pays tokens to children showing good behaviors as their rewards by allowing them to play/watch television. At-home devices such as television and computers are blockchain-connected and are accessed by children with sufficient tokens given by their parents when they do homework appropriately or help with housework. The smart contract features in Ethereum allow monitoring of children's activities through blockchain. Additionally, blockchain facilitates crowdfunding tertiary education by creating and storing "contracts in between students and their education sponsors facilitated by intermediary brokers denoted as fundraisers (Rashid et al., 2020, p.1)." The fund is organized and monitored by competitive fundraisers holding distributed ledgers and acting as miners in the blockchain.

5.3 Challenges and future directions

This section discusses the challenges that should be considered during educational blockchain implementation. Subsequently, based on the understanding of technological advances and frontier considerations and by considering the current research gaps and the challenges discussed, we suggest future research directions that need more attention to resolve these challenges and promote the future development of educational blockchain.

5.3.1 Validation, improvement, and extension

Currently, although with literature support, most blockchain applications in education remain as proposals without evidence for their effectiveness in the real world. Hence, their adaption to realistic educational scenarios should be tested before putting into practice. Such blockchain testing requires criteria depending on specific users. For instance, those responsible for online course payment may care about blockchain's throughput. Thus, it requires evaluating the average time between a transaction being sent and packed into the blockchain. Also, blockchain applications' effectiveness should be validated on actual learning data in large volumes. For example, Liang and Zhao (2020) indicate a need to optimize training programs, develop trustful educational systems, and stimulate educational reform via rational analysis of talent training based on massive education and talent data accumulated in their blockchain systems.

Furthermore, blockchain's general effectiveness in different contexts, courses, and institutions of different levels and among learners with different demographic characteristics (e.g., gender, culture, and high/low performance) should be validated. For example, Ubaka-Okoye et al. (2020) indicate a need to validate blockchain's potential on active e-learning platforms (e.g., mobile learning). Lam and Dongol (2022) suggest exploring smart contracts for different forms of assessments (e.g., peer assessments and self-assessments).

Another line of future work is blockchain systems' performance improvement by 1) finding appropriate blockchain versions, 2) trying novel frameworks (e.g.,

bespoke cryptocurrency for micropayments in e-learning), and 3) comparing proof-of-work and proof-of-authority consensus in the blockchain. For example, Rasool et al. (2020) suggest using image processing techniques to improve degree verification through institute logo identification and supporting colored degree documents. To facilitate accreditation authority, researchers suggest introducing a multisignature scheme (Turkanović et al., 2018) by distributing a single private key's power to multiple members acting as accreditation authority simultaneously.

Through validation and performance improvement, blockchain frameworks are expected to show high generalizability. For example, Bhagwat et al. (2020) plan to adapt their blockchain solution to more use cases like tender management, certificate distribution, and assignment incentivization. Turkanović et al. (2018) mention extending and adapting their grading framework to existing credit or certification systems. Tentea et al. (2019) indicate extending their assessment system by including different types of questions and creating classroom-like examinations.

Although we identified blockchain's popularity in various scenarios (e.g., management, storing, verification, and authentication of education documents, facilitating online learning and testing, and sharing education data and resources), there are other promising areas, including real-time awards, instruction evaluation, and supervision of students. First, by exploiting blockchain's currency property, instructors can use the "learning is earning" strategy (Sharples & Domingue, 2016) to promote students' motivation and encourage their participation in learning activities. According to smart contracts between teachers and students, real-time awards as digital currencies that can be used as tuition are offered to students via instructors' simple clicks. This strategy also applies to instruction evaluation by assessing teachers' teaching activities recorded in the blockchain and verifying teaching design and practice consistencies according to preplanned instructional activities submitted as smart contracts. Teachers meeting standards are given digital currency as rewards for their teaching skill improvement. Another promising area is supervision evaluation, where students' and supervisors' behaviors (e.g., how supervisors guide students) are recorded in blockchain ledgers for supervision evaluation according to supervisor responsibilities recorded in smart contracts.

5.3.2 Lack of knowledge and stakeholders' feedback

As blockchain is entirely new terminology, the technical complexity of cryptography and networking alongside complex concepts (e.g., primary key, public key, and recovery seeds) makes it difficult for users in education to understand. If blockchain is applied without proper know-how, training, and understanding, this complexity could unintentionally put institutions at risk for failure and a wasted investment (Hoy, 2017). Thus, educational practitioners and technicians ought to actively learn to understand and use blockchain. Our results indicate a trend for blockchain instruction through the project- and game-based learning to promote blockchain knowledge. In addition to students, blockchain instruction practice should also be applied to other involved stakeholders to promote the effective implementation of educational blockchain.

Attention should also be paid to blockchain usability improvement with easy interfaces for individuals lacking the technical expertise to easily understand and use. This requires case studies on end-users (e.g., students, parents, teachers, and

administrators) to evaluate system usability and improve its functionality by considering different respondent characteristics and feedback through dynamic assessment and repeating surveys. Throughout the different implementation and evaluation phases, constant consultation with stakeholders should be guaranteed to better understand, analyze, and address their needs and engage them to interact with each other to enable a trustable educational blockchain (Bessa et al., 2019).

5.3.3 Latency versus cost-efficiency

The high implementation, transaction, and computational costs hinder blockchain's application in traditional education systems (Alammary et al., 2019). The proof-of-work requirements in blockchain commonly require large amounts of "electricity, and the energy cost of a Bitcoin transaction can power 1.5 American homes for one day (Hoy, 2017, p.277)." Data redundancy also adds cost. This is because to manage and store massive student data, ledgers get longer, which increases algorithmic complexity and thus computational cost.

Thus, attention should reach to increasing cost efficiency during educational blockchain development and implementation via 1) step-by-step analysis of transformational measures and analysis of blockchain layers, including economic effect and transaction cost calculation; and 2) integrating and embedding blockchain into existing educational platforms/systems to reduce system complexity, cost, and work power caused by isolated storage of repetitive data (Srivastava et al., 2018). Other strategies include customized consensus algorithms to reduce network traffic and transaction time, estimate tradeoff between resource protection and flexibility based on applicable business models, and adopt cluster deployment to achieve server load balancing. Additionally, serverless cloud computing, a product of integrated computer and network technologies (e.g., distributed and parallel computing, network storage, and load balancing), can reduce blockchain deployment's time and labor costs based on the vertical resource-sharing concept (Rasool et al., 2020).

5.3.4 Ethics, policy, and regulation

In preserving individual privacy and autonomy, any solution must be European General Data Protection Regulation compliant. However, as blockchain eliminates the need for centralized authority, activity surveillance becomes extremely difficult. Additionally, since blockchain relies on dispersing information across a global network, international regulations could affect information shared across borders. Thus, adopting blockchain in education may face resistance. There are solutions (e.g., EvidenZ and BigchainDB), but any blockchain platform ought to cautiously consider international regulations. Additionally, it requires working with academic institutions/schools to understand internal regulations and ensure full compliance with these regulations (Hillman et al., 2019).

Besides compliance with existing regulations and laws, efforts on law and regulation establishment specifically for educational blockchain are needed. Since institutions may not intend to implement blockchain for all educational processes, they ought to define boundaries about how many processes of blockchain are to be

adopted (Alammary et al., 2019). Early collaborations between government and higher education with private sectors on regulatory compliance are expected to set the pace of educational blockchain applications. Further, attention should reach to personal privacy and intellectual property right protection during data storing and management, with clear declarations about who can access student data, what data to access, how data is protected, and what laws to protect. Although with cryptographic guarantees for transaction privacy, there will always be potential data leakage risks (e.g., frequent updates). Thus, efforts on leveraging Hyperledger Fabric to avoid these leakages are encouraged. Additionally, in data sharing, an agreement should be reached among all institutes about what data to share and how to share it.

5.3.5 AI and blockchain

Our results indicate an emerging trend toward using big data stored on the educational blockchain to leverage new AI paradigms like machine learning and deep learning. In return, AI helps develop machine learning systems on the blockchain with higher security, scalability, personalization, and governance (Dinh & Thai, 2018). For example, blockchain design and operation involve numerous parameters and tradeoffs between security, performance, and decentralization; AI eases these decisions, and automates and optimizes blockchain performance and governance. According to Kontzinos et al. (2019), appropriate data mining techniques and multi-criteria decision support algorithms, along with advanced visualizations, enable users to comprehensively understand their investigated data. For example, genetic algorithms analyze blockchain-based online examination processes throughout diverse task completion ranging from question generation to automatic marking/assessment (Wu, 2020). In Zhang et al. (2020a), autocorrelation quantitative feature analysis, fuzzy information clustering, and the root game equilibrium optimization algorithm allows high-quality extraction and adaptive optimization control of educational resources based on blockchain-trusted big data. Additionally, there are practices on integrating blockchain, face recognition technologies, AI, virtual reality, whole-body motion capture technologies, and laser optical positioning to create new virtualized and immersive learning experiences, enrich classroom-teaching contents, and create open learning without space and time constraints. We summarize the potential improvement of educational blockchain brought by AI as follows.

1) Security

   The fact that smart contracts may involve bugs or faulty programming codes makes them vulnerable to crashes and attacks and hinders their application in education. According to the Gartner 2019 CIO Survey, 47% of education respondents lack confidence in blockchain and take a wait-and-see approach to determine its potential risks.[4] Thus, it requires detecting bugs in smart contracts to realize intelligent quality assurance. Advanced machine learning combined with visualization technologies can help promote intelligent quality assurance by monitoring blockchain systems' performance, detecting vulnerable programming codes in smart contracts, identifying risks and

[4] https://www.gartner.com/smarterwithgartner/4-ways-blockchain-will-transform-higher-education/

attacks in transactions, and automatically invoking appropriate defense mechanisms in real-time mode (Dinh & Thai, 2018). Accordingly, proactive and autonomic actions can then be generated to prevent disruptive and illegal actions in blockchain systems.

2) Scalability

Massive users may request smart services on campus; however, blockchain's scalability problem (i.e., a blockchain transaction is slow) (Alammary et al., 2019) brings difficulties in adapting to massive users simultaneously. AI allows a more scalable and robust blockchain by creating blocks at a faster speed to promote throughput when transactions surge (Dinh & Thai, 2018). For example, to enhance scalability and increase throughput, researchers can consider using deep reinforcement learning to regulate the block producers, consensus algorithms, block sizes, and block intervals in a dynamic mode based on quantitative evaluation of the blockchain system's scalability, decentralization, security, and delay (Liu et al., 2019).

3) Personalization

Our results indicate an emerging trend in adaptive education and personalized learning facilitated by blockchain, mainly through individualized competence development, education curriculum customization, and solving cold-start problems in learner data analytics. Specifically, blockchain solves cold-start problems during learning analytics by constantly tracking learning achievements (Ocheja et al., 2019) and accessing learning logs and events (e.g., a register of property rights) of new learners from other institutions. The data stored on the blockchain can be used to analyze individuals' talents and interests to automatically develop competencies tailored to their purposes, strengths, and characteristics via curriculum personalization (Lam & Dongol, 2022). Blockchain also allows learners to trade their learning results in cryptocurrencies, which in turn gives them strong motivation to acquire knowledge independently to gain rewards.

Coupled with AI, internet resources, and network technologies, blockchain provides privacy and personalization simultaneously (Dinh & Thai, 2018). A decentralized content provider (e.g., a social network on the blockchain) leverages AI on students' sides to personalize learning materials. The use of machine learning helps evaluate students' browsing behaviors and hobbies based on blockchain data analysis to promote instructors' provision of suitable, personalized learning materials to individuals. Since the whole computation is conducted locally, no personal data will leave students' devices, thereby ensuring privacy. Additionally, student content preference sanitization prevents content providers from profiling users.

6 Conclusion

The present study is the first in-depth to track current advances in educational blockchain research using topic modeling-based bibliometrics. Such work is timely as educational blockchain is attracting intense attention from academia, industries, governments, and educational institutions. Our results indicate blockchain's prevalence in online testing and learning, education data and resource sharing, educational record verification and authentication, data management and storing, administration

management, education/discipline of children, and crowdfunding tertiary education. Blockchain also facilitates LA, personalization, and high-quality education resource extraction resolves the drawbacks of college curriculum non-transparency and centralization, and prevents information falsification and leakage in admission applications. Technologically speaking, little effort is spared in integrating advanced technologies like machine learning with blockchain. We also reveal challenges of educational blockchain, including validation, improvement, extension, latency, stakeholders' lack of knowledge, and issues about ethics, policy, legal, scalability, and security. We thus highlight the need to 1) validate systems across diverse application scenarios and enhance their generalizability; 2) educate stakeholders about blockchain and its applications via game- and project-based learning; 3) obtain feedback from stakeholders to optimize blockchain systems' functionality and design and bring them together to promote trustable educational blockchain use; 4) integrate blockchain solutions into existing educational systems and exploit serverless cloud services to improve cost-efficacy; 5) comply with international and institutions' regulations and work closely with stakeholders on law and regulation establishment; and 6) exploit AI to enhance scalability, security, justification, and personalization through analyzing big data accumulated in blockchain systems to capture abnormal behaviors, automate decisions making, and understand users' behaviors and preferences.

Data availability statement The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interests The authors declare that they have no conflict of interest.

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